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# Systematics of Paleomagnetic Directions from Early–Middle Devonian Rocks of Minusa Troughs: New Data and Old Problems

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**Abstract**—Paleomagnetic results from numerous Early–Middle Devonian volcanic sequences of the Minusa trough, southern Siberia, are presented. The analysis of these data definitely indicates that the geomagnetic field in the Devonian had a specific character, different from both the present field and the field of more ancient geological epochs, and was extremely variable (hyperactive). The anomalies in the paleomagnetic record of the Early–Middle Devonian are not local, peculiar to a particular region, but have a global occurrence. The synthesis of the obtained results with the paleomagnetic data from the coeval volcanics from Scotland shows that during a relatively short time (10–20 Ma), the geomagnetic pole repeatedly changed its location, significantly deviating from the Earth's rotation axis up to the (paleo)equator and assuming some quasi-stable positions. The arguments suggesting that the specific features of the Devonian paleomagnetic record could probably be induced by the significant contribution of the equatorial dipole to the main geomagnetic field are presented.

*Keywords:* the Devonian, paleomagnetism, equatorial geomagnetic dipole, axial geomagnetic dipole, Siberian platform, true polar wander

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## INTRODUCTION

The problem of the Devonian paleomagnetism dates far back to the beginning of the paleomagnetic research when this problem was noted by K. Creer, who studied the Lower Devonian Old Red Sandstone (ORS) volcanic–sedimentary formation of British Caledonides (Creer and Embleton, 1967 and references there). Initially, this problem only consisted of the presence of two significantly different paleomagnetic directions recorded in the ORS rocks. These directions, which were referred to as DI and DII in the literature (Creer and Embleton, 1967), were detected in the same sections and yielded significantly different paleolatitudes. A similar situation was revealed by the Russian researchers in the European part of the territory of the former Soviet Union (*Paleomagnetism*..., 1974). The discussion on the nature of the DI and DII directions resulted in the view, shared by most researchers, according to which the DI direction came to be considered as the result of the Late Paleozoic remagnetization. This interpretation was induced by the fact that, on one hand, the paleomagnetic pole corresponding to this direction was close to the Carboniferous–Permian poles and, on the other hand, the DII direction was more consistent with the paleoclimatic zonality for the Devonian of the East European Platform (Creer, 1968; *Paleomagnetism*..., 1974). At the same time, the ORS studies also revealed other quasi-stable paleomagnetic directions, which are periodically repeated in the sections and which were interpreted as the transition zones with respect to the DII directions of direct and reverse polarity (Sallomy and Piper, 1973; Kono, 1979).

Just as many problems also appeared in the paleomagnetic studies of the Devonian volcanics from the Asian part of the former Soviet Union (here, we will not specially focus on the old results that were obtained with the use of low-intensity and low-detail demagnetization). In particular, the studies of the Late Devonian–Early Carboniferous kimberlites and traps of the Siberian platform (Kravchinsky et al., 2002; Konstantinov and Stegnitskii, 2012; Orlov and Shatsillo, 2011; Shatsillo et al., 2014; etc.) have clearly shown that the range of the paleomagnetic directions contained in these objects is not limited to the DI and DII directions in Creer's terminology but contains many other anomalous directions (including the systematic ones) whose existence cannot be explained by the local tectonics or by the hypothesis of the remagnetisation. Furthermore, it turned out that the number of the anomalous Devonian paleomagnetic directions obtained from these objects is commensurate with the number of the expected[1] ones, which unequivocally

[1] Based on the position of the respective paleomagnetic poles relative to the Paleozoic segment of the apparent polar wander path (APWP) curve for the Siberian platform.

suggests some specificity of the geomagnetic field in the Devonian. In fact, a similar result was obtained from the Late Devonian lavas of the Northern Tien Shan (Bazhenov and Levashova, 2011): the authors attribute the anomalies fixed in the paleomagnetic record to the secular variations of the geomagnetic field and, based on the example of the studied objects, conclude that the amplitude of secular variations is not constant in time. In any case, the set of the paleomagnetic data accumulated to date for the Devonian rocks strongly suggests that the behavior of the field in the Devonian was significantly different from both the present field and from the field of the other geological epochs. For studying the Devonian geomagnetic phenomenon and developing the corresponding dynamo models, first of all, new paleomagnetic data are needed.

In this work, we present new paleomagnetic results from the Early–Middle Devonian objects of the Minusa troughs, their systematics, and comparison with the coeval volcanics from the ORS formation of British Caledonides.

The fact that valid paleomagnetic data for Early–Middle Devonian of the Siberian platform are practically absent attaches additional relevance to this study. Obtaining these data in the conducted works would make it possible to solve a range of geodynamical and tectonic questions associated with the Paleozoic evolution of the structures of North Eurasia.

## GENERAL INFORMATION ABOUT THE GEOLOGY OF THE REGION AND OBJECTS OF STUDY

The object of this study was the Early–Middle Devonian volcanogenic formations composing the lower part of the Minusa troughs sedimentary sequence. According to the existing data on the tectonics of the region (Zonenshain et al., 1990; Berzin et al., 1994), the Minusa troughs were formed on the heterogeneous basement of the south-Siberian Caledonides, which constituted a single structure with the craton part of Siberia since the end of the Ordovician. The Minusa troughs are a series of isometric synclines (centroclines) separated by the anticlinal uplifts and include the South Minusa, Syda-Yerba, Chebaki–Balakhta (North Minusa), and Nazarovo troughs (Fig. 1). The studied volcanic sequences have a bimodal composition (rhyolites, dacites, basalts and their tuffs) with alkaline specificity. Within the Minusa troughs, the Devonian volcanics are laterally variable, both by the ratio of the acidic and basic varieties and structural–textural features and by thickness. According to this variability, a number of the formations are distinguished (Immir, Ton, Chilan, Kharadzhul, etc.), which are combined into the Byskar group within most of the region (*Geologicheskaya*..., 1957; 1959). The sedimentary units within the Byskar group are strongly subordinate relative to the volcanic formations. The Byskar group overlies the subjacent formations of different ages with angular unconformity, erosion, and conglomerates at the base. The thickness of the Byskar group in its most complete sequences reaches 1.5–2 km. The Byskar layers are concordantly and with erosion overlain by terrigenous-carbonate variegated units with a total thickness ~0.5 km, containing in their upper part (the Beya Formation) pelecypods, brachiopods, and bryozoans characteristic of the upper parts of the Givetian faunal stage of the Middle Devonian (*Geologicheskaya*..., 1957) and Eifelian-Lower Givetian flora at the base (the Toltak Formation) (*Geologicheskaya* ..., 1959). The section of the Minusa troughs is completed with the Late Devonian red rocks and Tournaisian and Visean volcanic-sedimentary strata, which are erosionally overlain by the Early Permian coal-bearing formations. A number of the modern geochronological determinations are obtained from the Byskar volcanics (Ar-Ar and U-Pb ages reported in (Vorontsov et al., 2012)), which fall in the interval $407.5 \pm 0.2$ to $386 \pm 4$ Ma; younger datings are obtained from the dikes crosscutting the volcanic units ($385 \pm 4$ and $364 \pm 5.5$ Ma). Thus, the existing geochronological and paleontological data constrain the time of accumulation of the main volume of Byskar volcanics to the Emsian and Eifelian, which covers an interval of ~20 Ma.

During the field studies of 2010–2012 within the Minusa troughs, we tested the main sections of the Byskar volcanics (the samples were taken only from the sections where it was possible to establish the primary bedding of the strata). The study covered the sections in the southwestern (along the Abakan and Dzhebash rivers), eastern (Tuba River), and northern parts (Koksa Bay, Krasnoyarsk Reservoir) of the South Minusa trough; the south of the Syda-Yerba trough (Krasnoyarsk Reservoir, upstream of the Bellyk village); and southeast (Sisim Bay), north (Krasnoyarsk Reservoir, downstream of the Ezagash Bay, the Truba (Pipe) section), and the central part (Krasnoyarsk Reservoir, in the vicinity of Novoselovo village) of the Chebaki–Balakhta trough (Fig. 1). The studied sections represented the fragments of the limbs of the large folds or the monoclines with a consistent bedding attitude within the each section. As the attitude elements, the dip angles of the overlying sedimentary strata, the rare sedimentary interbeds within volcanics, and the planes of magmatic parting corresponding to the boundaries of the lava flows were used. Paleomagnetic samples were acquired by the site sampling procedure according to which 6–10 oriented samples over an area of 1–3 m were taken from each magmatic body. A total of 118 sites representing separate lava flows or subvolcanic bodies, or, in rare cases, sedimentary interbeds among volcanics were examined.

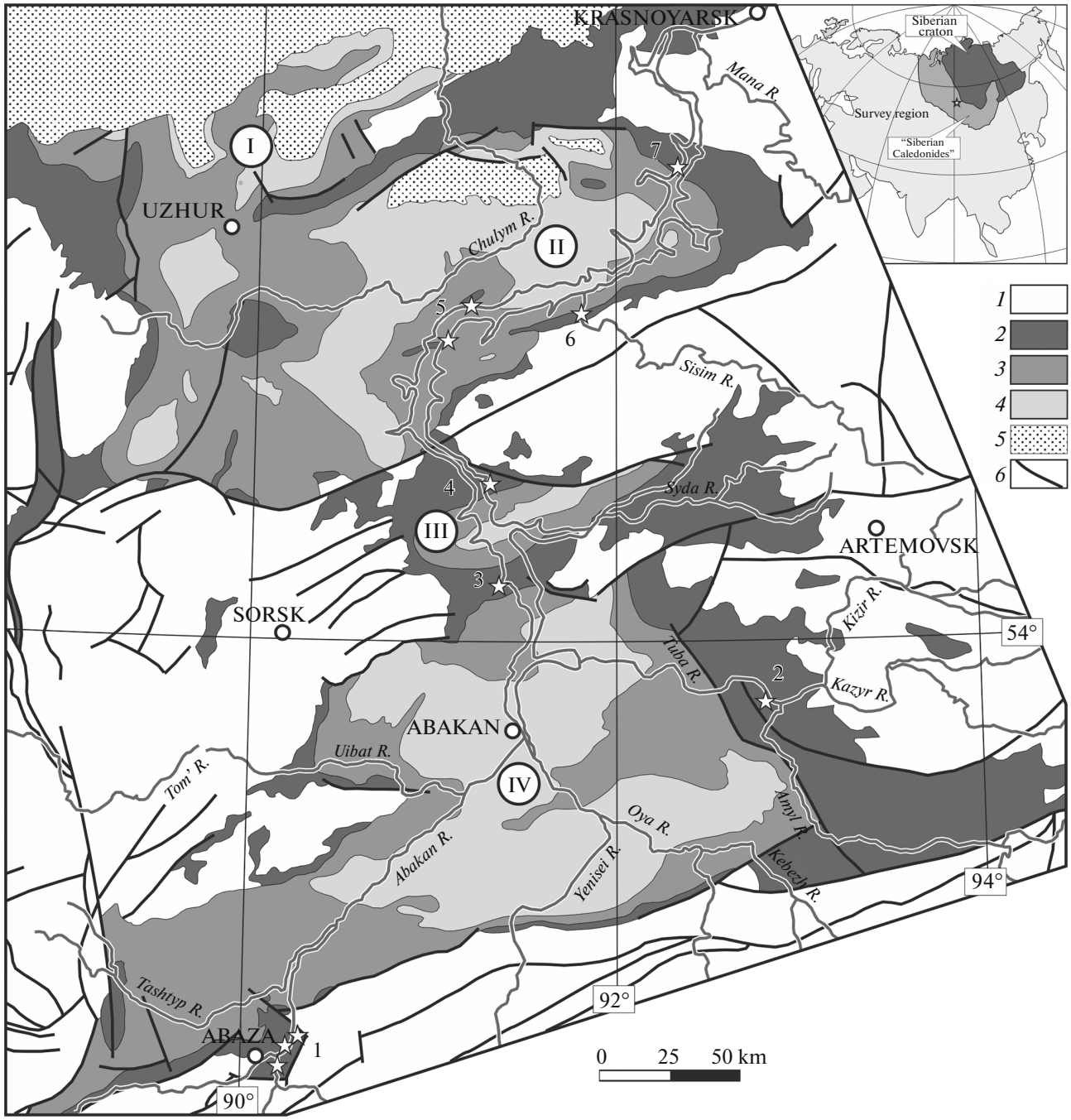



**Fig. 1.** Schematic geological structure of Minusa trough and positions of study objects. Based on 1 : 2500000 Geological Map of the former Soviet Union (Nalivkin D.V., Ed., 1983). *1*, pre-Devonian basement; *2*, Lower–Middle Devonian volcanic–sedimentary formations; *3*, Middle–Upper Devonian; *4*, Carboniferous–Lower Permian; *5*, Jurassic and younger sediments; *6*, tectonic dislocations. Roman numerals in circles indicate depressions of Minusa trough: I, Nazarovo; II, Chebaki–Balakhta; III, Syda-Yerba; IV, Minusa. Asterisks and respective numbers are studied sections: *1*, along Dzhebash and Abakan rivers; *2*, Tuba; *3*, Koksa; *4*, Bellyk; *5*, Novoselovo; *6*, Sisim; *7*, Truba.

## RESULTS OF LABORATORY STUDIES AND THEIR INTERPRETATION

The collections of samples were processed in the laboratories of the Schmidt Institute of Physics of the Earth (IPE) of the Russian Academy of Sciences (RAS) and Geological Institute (GIN) RAS. All the samples were subjected to detailed thermal demagnetization until the magnetization completely diasppeared; the remanent magnetization was measured on AGICO JR-4 and JR-6 spinner magnetometers (Czech Republic). The results of the measurements

were processed with the use of the software package developed by R. Enkin (Enkin, 1994); the cluster analysis of the distribution of the vectors on the sphere was conducted with the use of the SteroNett program v2.46 (Johannes Duyster).

The studied rocks have broadly varying magnetic characteristics and the content of magnetization components which differ both among different sections and, frequently, among different sites within one section. Due to this, it is impossible to present all the types of the Zijderveld diagrams in the scope of one paper. Therefore, below we mainly focus on the cluster analysis of the distribution of the site-mean magnetization directions. We will show the most representative Zijderveld diagrams for the revealed groups of predominant directions (clusters).

The site mean directions were mainly calculated from the end (the most stable) high-temperature magnetization components; in some cases, for calculating the mean directions, we also used the "intermediate" medium- and low temperature components, as well as the remagnetization circles, although only if the latter were coherent with the end components. Other low- and medium-temperature intermediate components were excluded from the analysis.

Since for the segments with strongly developed fault-block tectonics (the sections on the Abakan and Dzhebash rivers), we assume local rotations, the results for these objects will be considered separately from the other part of the data.

### *Sections along the Abakan and Dzhebash Rivers*

The section along the Dzhebash River (the right tributary of the Abakan River, ~7 km downstream of the town of Abaz) is an extensive monocline dipping southwest (326°–335°) at an angle ranging from 40° to 53°. In this section, 20 remote sites from different stratigraphic levels were sampled on both sides of the river valley. The distance (across the strike) between the remotest sites is ~2 km. In 15 of the studied sites, it was possible to isolate the magnetization components (Table 1, Fig. 2). The means for nine sites of the Dzhebash section form a compact group whose fold tests ((DC, W&E (Enkin, 2003; Watson and Enkin, 1993)) suggest post-folding magnetization; the other directions are distributed chaotically. The post-folding magnetization component of the Dzhebash section is close to the expected direction of Carboniferous–Permian remagnetization (D = 292.1°, I = –63.5°, recalculated into the coordinates of the studied region), which we previously revealed in the Late Devonian and Early Carboniferous rocks of the Minusa troughs (Shatsillo et al., 2012) (Fig. 2, inset). At the same time, the difference in the declinations (~20°) for these directions, which have close inclinations can indicate a certain clockwise rotation of the Dzhebash block in the post-Paleozoic time. The direction identified in sites 048-2 and 049 (combined) is close to the expected Devonian direction for the region. The Zijderveld diagrams for the samples from the Dzhebash section, which characterize the post-folding direction and the expected Devonian direction, are shown in Fig. 3. We note that most samples containing the post-folding magnetization component have a fairly noisy signal: the magnetization vector during the demagnetization wanders around a given direction, whereas under heating above 580–590°C, it starts to chaotically change direction.

Along the Abakan River, we sampled 11 sites on two remote areas (in the left slope of the valley downstream of the Kharadzhul River mouth and in the right slope about 3 km upstream of the mouth of the Bol'shoi Monok River) with substantially different attitude elements. All the sampled sites contain an interpretable paleomagnetic signal. The analysis of the site mean directions reveals two clusters[2] on the stereogram (Table 2, Fig. 2). One of them, cluster I, which includes four sites, is identified by the fold tests (DC, W&E) as pre-folding. The fold tests for the second cluster (II), which includes three sites, do not give a clear result, which is due to the insignificant difference in the bedding elements of the sites. The mean directions for the other sites are distributed quasi-chaotically. The Zijderveld diagrams for the samples characterizing clusters I and II are shown in Fig. 3.

As in the case of the post-folding component of the Dzhebash section, the mean direction for the pre-folding Abakan cluster I coincides by inclination with the direction of the Carboniferous–Permian remagnetization of the region (Fig. 2, inset); however, the difference in the declinations suggests the counterclockwise post-Permian rotation of the Abakan block (~60°). We interpret the considered pre- and post-folding directions of the Dzhebash and Abakan sections as the result of the simultaneous Carboniferous–Permian remagnetization. This implies the following tectonic history for the studied areas: (1) deformations of the Dzhebash block; (2) remagnetization; (3) deformations of the Abakan block; and (4) differently directed rotations of the Dzhebash and Abakan blocks. This conclusion is supported by the fold test (New Fold Test, NFT) (Shipunov, 1995) for the combined set of the pre- and post-folding components which yields a stable solution indicating the presence of two magnetization components that were formed during the deformations. The mean inclination calculated for the combined set of the metachronous component (in the coordinate systems of the magnetization formation) is $I = -65.1 \pm 2.8$ at $n = 13$, $k = 120.9$. We note that with partial correction for the rotation of the Abakan block, cluster II becomes close to the NWSE cluster revealed from a number of the other Minusa objects which will be discussed below; the fold test,

[2] By cluster we mean a dense group that includes at least three directions.

**Table 1.** Paleomagnetic directions contained in the volcanics of the Dzhebash river section (site means)

| No. | Site | SLONG | SLAT | *n* | Dg | Ig | Ds | Is | *k* | a95 | *T*1 | *T*2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 045-1 | 90.216 | 52.640 | 4 | 295.9 | –60.9 | 203.5 | –65.7 | 253.1 | 5.8 | 521–580 | 638–653 |
| 2 | 045-2 | 90.216 | 52.640 | 5 | 323.1 | –30.9 | 305.5 | –68.9 | 15.4 | 20.1 | 400–503 | 503–580 |
| 3 | 045-3 | 90.216 | 52.640 | 6 | 305.4 | –58.6 | 206.8 | –70.9 | 137.4 | 5.7 | 260–592 | 503–653 |
| 4 | 045-6 | 90.216 | 52.640 | 5 | 293.7 | –64.2 | 196.3 | –64.1 | 24.6 | 15.7 | 400–576 | 503–592 |
| 5 | 045-7 | 90.216 | 52.640 | 3 | 332.6 | –69.3 | 157.6 | –70.7 | 38.7 | 20.1 | 400–459 | 459–576 |
| 6 | 045-8 | 90.216 | 52.640 | 5 | 315.0 | –61.8 | 190.7 | –73.9 | 36.3 | 12.9 | 260–459 | 576–580 |
| 7 | 045-9 | 90.216 | 52.640 | 3 | 280.1 | –71.9 | 183.1 | –57.5 | 29.0 | 23.3 | 260–482 | 400–567 |
| 8 | 045-10 | 90.216 | 52.640 | 3 | 327.5 | –68.0 | 164.0 | –71.7 | 30.0 | 22.9 | 400–482 | 503–576 |
| 9 | 045-11 | 90.216 | 52.640 | 6 | 135.7 | –76.6 | 149.5 | –37.2 | 53.4 | 9.3 | 400–459 | 521–592 |
| 10 | 048-1 | 90.210 | 52.632 | 6 | 313.4 | –71.2 | 153.0 | –55.2 | 21.6 | 14.7 | 342–459 | 503–576 |
| 11 | 048-2 | 90.210 | 52.632 | 4 | 334.3 | –60.3 | 135.9 | –66.1 | 18.9 | 21.7 | 400–459 | 503–580 |
| 12 | 049+048-2 | 90.212 | 52.633 | 7 | 351.4 | 11.3 | 355.0 | –33.6 | 27.8 | 11.7 | 260–432 | 567–653 |
| 13 | 050 | 90.196 | 52.642 | 4 | 279.3 | –45.7 | 221.7 | –52.0 | 28.2 | 17.6 | 432–482 | 482–580 |
| 14 | 051 | 90.195 | 52.643 | 3 | 174.4 | 70.6 | 319.9 | 61.5 | 152.5 | 10.0 | 432–521 | 503–580 |
| 15 | 053 | 90.195 | 52.648 | 4 | 101.4 | 7.3 | 86.5 | 31.5 | 20.4 | 20.8 | 432–459 | 567–580 |
| MEAN for nos.1, 3–8, 10, 11 | | | | 9 | **311.2** | **–66.1** | | | **80.6** | **5.8** | | |
| | | | | | | | 175.9 | –68.0 | 48.1 | 7.5 | | |

No. is direction number; site is site number; SLONG/SLAT is geographic longitude and latitude of sampling site; *n* is number of samples or sites included in statistics; D/I is declination/inclination (indices s/g denote stratigraphic/geographic coordinate system); *k* is precision parameter; a95 is radius of confidence circle corresponding to probability 95%; *T*1/*T*2 are initial/final temperatures of isolation of magnetization components.

**Table 2.** Paleomagnetic directions contained in volcanics of Abakan river section (site means)

| No. | Site | SLONG | SLAT | *n* | Dg | Ig | Ds | Is | *k* | a95 | *T*1 | *T*2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 054-1 | 90.279 | 52.732 | 6 | 298.2 | –1.5 | 284.4 | –19.2 | 96.1 | 6.9 | 432–580 | 521–592 |
| 2 | 054-2 | 90.279 | 52.732 | 4 + 1c | 297.5 | 5.3 | 290.6 | –15.9 | 43.5 | 12.1 | 459–538 | 567–622 |
| 3 | 054-3 | 90.279 | 52.732 | 5 | 291.2 | 6.8 | 289.3 | –9.6 | 40.3 | 12.2 | 459–521 | 503–580 |
| 4 | 055 | 90.282 | 52.735 | 6 | 203.7 | 5.9 | 235.2 | –65.5 | 193.1 | 4.8 | 432–459 | 503–580 |
| 5 | 056 | 90.281 | 52.734 | 4 | 214.5 | 3.0 | 252.7 | –57.8 | 644.0 | 3.6 | 260–459 | 567–592 |
| 6 | 057-1 | 90.289 | 52.826 | 4 | 332.4 | –33.7 | 223.5 | –69.1 | 12.1 | 27.5 | 432–459 | 521–592 |
| 7 | 057-2 | 90.289 | 52.826 | 4 | 327.2 | –41.4 | 213.3 | –61.6 | 63.0 | 11.7 | 432–459 | 567–592 |
| 8 | 057-3 | 90.289 | 52.826 | 6 | 12.7 | 24.5 | 16.5 | –39.3 | 14.9 | 17.9 | 482–564 | 580–638 |
| 9 | 058-1 | 90.297 | 52.835 | 5 | 194.8 | 48.6 | 328.2 | 67.4 | 467.2 | 3.5 | 400–482 | 604–638 |
| 10 | 058-2 | 90.297 | 52.835 | 6 | 185.2 | 47.9 | 346.5 | 78.7 | 171.7 | 5.1 | 400–564 | 638–680 |
| 11 | 059 | 90.295 | 52.841 | 6 | 3.2 | 37.6 | 12.9 | 29.0 | 91.3 | 7.0 | 400–445 | 638–680 |
| MEAN for cluster I nos.4–7 | | | | 4 | 258.2 | –30.6 | | | 1.6 | 125.1 | | |
| | | | | | | | **232.2** | **–64.3** | **76.8** | **10.5** | | |
| MEAN for cluster II nos. 1–3 | | | | 3 | 295.6 | 3.5 | | | 191.4 | 8.9 | | |
| | | | | | | | 288.1 | –14.9 | 195.4 | 8.8 | | |

Index c in column *n* denotes great circles (of demagnetization) used for calculating site mean direction; other explanations are presented in notes to Table 1 and in text.

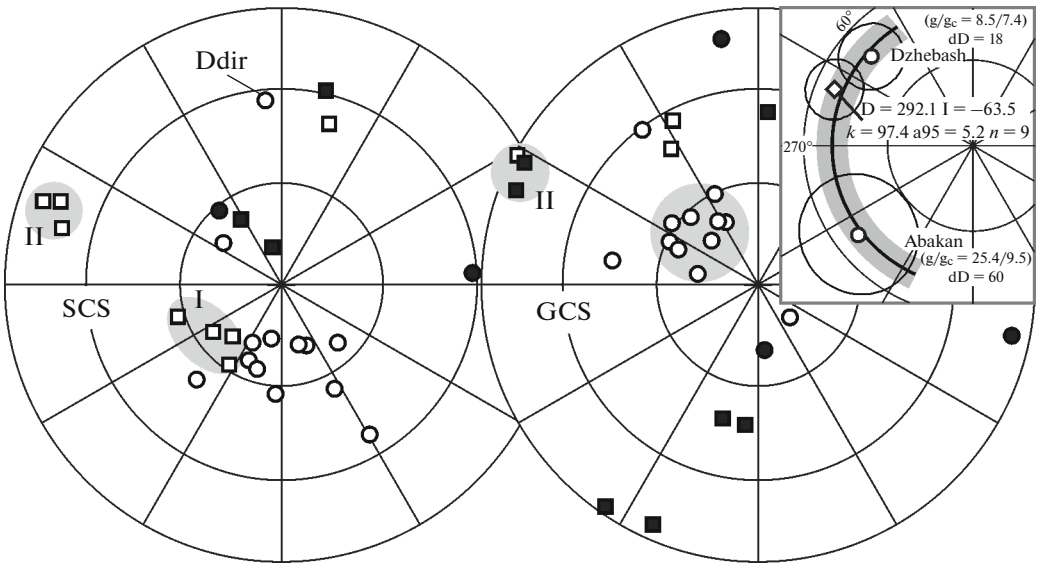



**Fig. 2.** Distribution of site-mean directions for sections along rivers Dzhebash (circles) and Abakan (squares). Gray filled areas are groups of vectors (clusters I and II) discussed in text; Ddir is site containing expected Devonian direction. Hereinafter, GCS/SCS are geographical/stratigraphic coordinate systems; filled and open markers are projections on lower and upper hemisphere. Inset: comparison of mean directions of metachronous magnetization component of Dzhebash and Abakan sections with direction of Late Paleozoic remagnetization for Minusa trough (Shatsillo et al., 2012). Diamond marks direction of Late Paleozoic remagnetization for study area; heavy gray line is mean inclination of metachronous component with allowance for confidence interval calculated for combined data from Dzhebash and Abakan sections. Angular distances (in degrees) between directions of metachronous component and direction of Late Paleozoic remagnetization along great-circle arc (g) and its critical value ($g_c$) and declination difference (dD) are shown in parentheses. Other explanations are in text.

which only analyzes inclinations (Enkin and Watson, 1996) when applied to the combined set (NWSE + II), suggests that these directions are pre-folding.

### *Systematics of Paleomagnetic Directions from the Tuba, Koksa, Bellyk, Novoselovo, Sisim, and Truba Sections*

The separation of the paleomagnetic directions revealed in the Devonian rocks of these objects causes no less difficulty: although the directions are reasonably densely grouped within the site volumes, the site-to-site scatter even within a single section is significant (Fig. 4, Table 3). In contrast to the objects of the southwest of Minusa (the Dzhebash and Abakan sections), the considered areas are tectonically relatively quiet, which allows us to consider the data for these areas jointly. We note that within the volume of a single section, the attitude elements of the bedding are very close to each other and, hence, it is impossible to conduct the fold test within the volume of the section; therefore, these data are not presented in this paper. The analysis of the entire set of site directions reveals a number of more-or-less statistically valid clusters (Fig. 5e, Table 4).

Part of the directions does not form any groups at all and, in the first approximation, is chaotically distributed on the sphere. These directions, which are conditionally related to type I, are mainly detected in the Sisim Bay section, which is a monocline and in the Koksa section (south) representing the southern limb of a large anticline. The contribution of the directions of type I relative to the entire set of directions is ~9%.

In type II we classify the directions that are only repeated within the volume of a single section. This type includes clusters E and SW characterized by the directions from the Novoselovo-2 section representing the northeastern limb of a large anticline and the direction from the Koksa section (south), respectively. About 10% of the directions of the entire set pertain to type II.

Finally, type III comprises the groups of the directions that are systematically repeated in the remote sections. This type includes clusters N (~24%), NW (~6%), NWSE (~9%), P (~7%), and S (~35%). As seen in Figs. 5a–5d, the considered clusters become differentiated from each other when considered in the stratigraphic coordinate system, whereas in the geographical coordinate system they become spread out and any additional groups of the directions do not appear. It is important here that all the revealed directions differ from the known post-Devonian directions for Siberia. Cluster P is the exception: with confidence interval taken into account, this cluster is located relatively close to the expected directions for the Jurassic (Fig. 5f). The fold test (DC) indicates pre-folding magnetization for all the clusters except for cluster S where the test is unconclusive. The Zijderveld diagrams characterizing all the described directions are shown in Fig. 6.

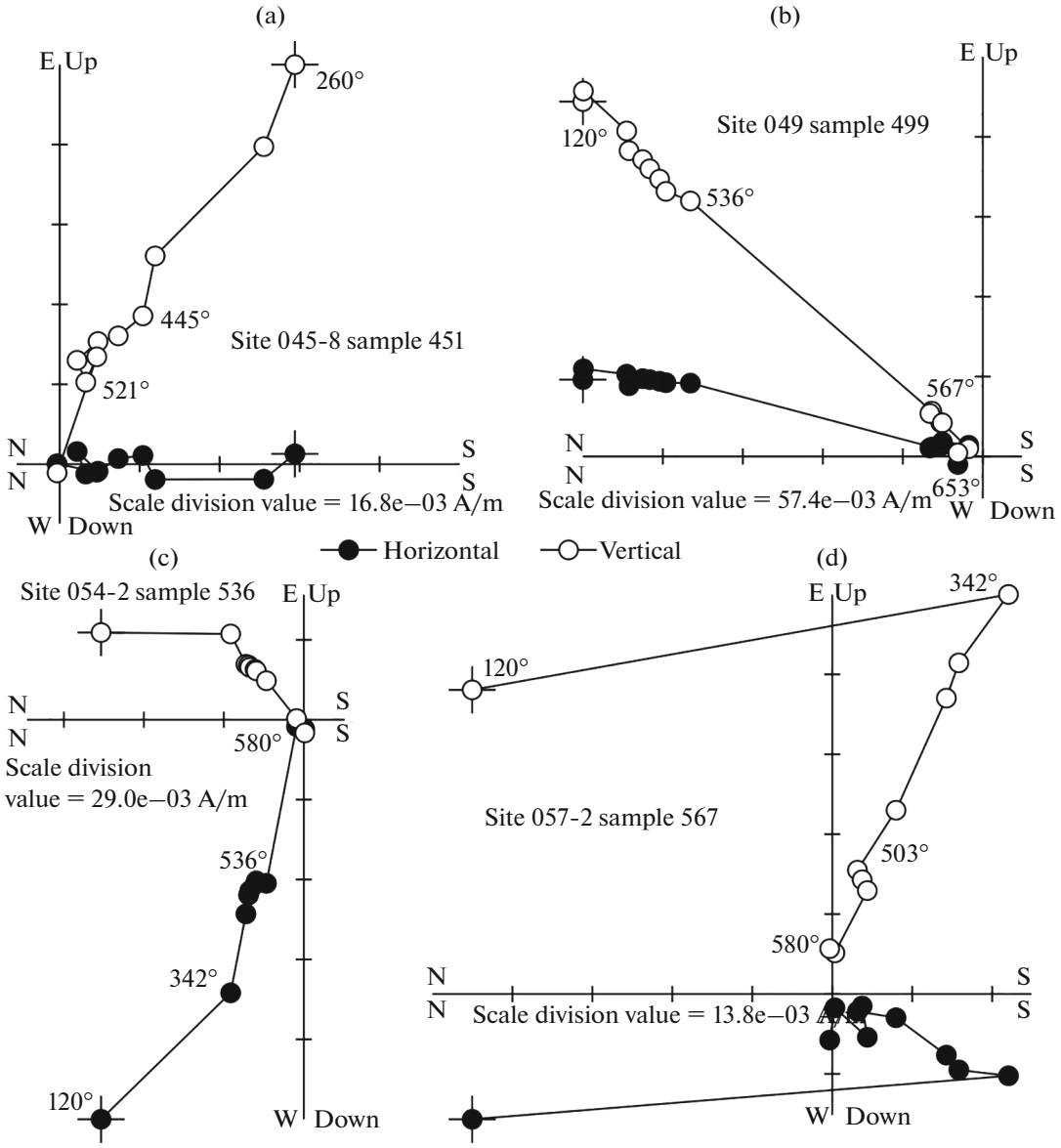



**Fig. 3.** Examples of thermal demagnetization of Devonian volcanics from sections along (a, b) Dzhebash river and (c, d) Abakan river. (a) Post-folding component; (b), expected Devonian direction; (c), pre-folding component of cluster I; (d), component of cluster II. Zijderveld diagrams are shown in stratigraphic coordinate system.

Here, a question on the meaning of the observed character of the paleomagnetic record in the studied Devonian rocks naturally arises. At the current stage of the paleomagnetic analysis, we can make the following hypothetical suggestions.

Perhaps the simplest explanation of the directions pertaining to types I and II lies in the assumption that these nonsystematic directions are the result of recording the magnetic field during the reversal or excursion or, in the case of directions of type I, they can be a certain irregular mixture of components having different directions and different ages. The ubiquitously present S and N directions are closely antipodal and successfully pass the reversal test ($\gamma/\gamma_c = 10.5/11.0$). It is these directions that are close to the expected Devonian directions of Siberia and are interpreted by us as the Devonian itself, i.e., reflecting the positions of the Earth's rotation axis in the Devonian. The systematic anomalous (for the Devonian) directions of type III corresponding to clusters NW, NWSE, and P are probably most enigmatic. If direction P can formally be considered the result of the pre-folding Jurassic remagnetization,[3] while NW is poorly statistically valid, then the NWSE direction, with the results from the sections of the

[3] Occurrence of deformed piles of Lower-Middle Jurassic within the Chebaki–Balakhta trough is consistent with this claim (Bezzubtsev et al., 2000), however it's also well known that main post-Caledonian deformation within the limits of the Central-Asian Fold belt took place in the Late Paleozoic (Zonenshain et al., 1990; Berzin et al., 1994).

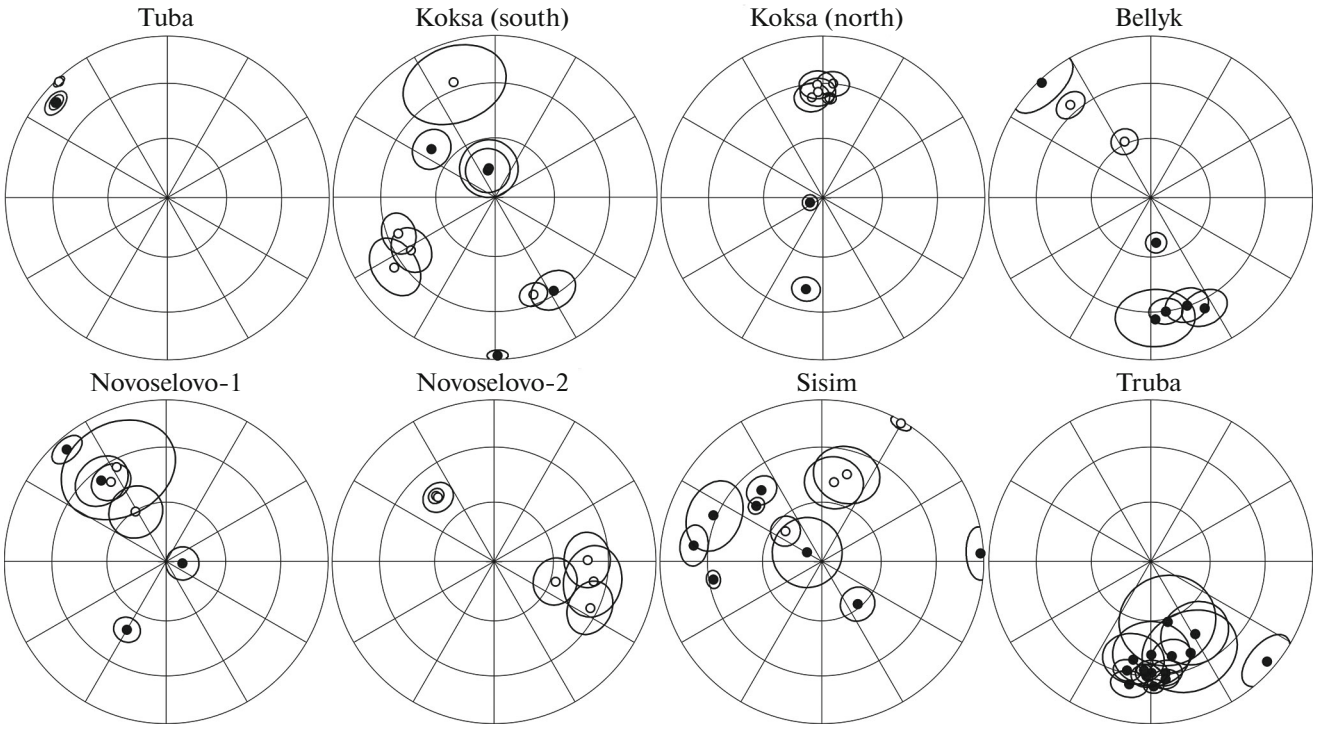



**Fig. 4.** Distribution of site-mean paleomagnetic directions for Tuba, Koksa, Bellyk, Novoselovo, Sisim, and Truba sections. All stereograms are shown in stratigraphic coordinate system. Other explanations are in Fig. 2.

Abakan River taken into account, can hardly be understood as an artifact. The coexistence of the NWSE direction with the S and N directions must imply the occurrence of some quasi-stable non-axisymmetric states of the field that were repeated in time and were recorded in this direction.

There is probably a certain system in the distribution of the anomalous directions of type III (NW, NWSE, and P): the poles calculated from these directions perfectly fall in the great-circle arc which also passes through the mean coordinates of the study region (Fig. 5f). Should the Devonian directions also fit in this system, this regularity could indicate that the anomalous directions are the result of the significant and time-varying contribution of the zonal non-dipole (quadrupole and octupole) components into the main geomagnetic field, which distort the inclinations. However, this model is actually not valid and the expected Devonian directions (S and N), just as the directions of type II (E and SW) do not fit in this system. Another probable explanation of the observed relationship between the NW, NWSE, and P directions suggests that the NW direction is a superposition of the anomalous primary NWSE direction and the pre-folding metachronous P direction which has the Mesozoic age. However, this model requires the assumption that the contamination of the NWSE direction by the P direction was identical in the remote sections, which is unlikely.

In any case, the obtained data indicate that about half of the studied time interval (in terms of the proportion between the "expected" Devonian and other directions) was occupied by the excursion–reversal or some other dynamo processes under which the geomagnetic pole significantly deviated from the field of the central axial dipole. Due to the problems of the regional correlation of the studied sections it is impossible to elucidate the pattern of the proportion of the revealed direction in the overall stratigraphic sequence. However, the data within an individual section rather indicate that the change of these directions occurs in a complicated, probably non-systematic way (Table 5). It is also worth noting that the studied sections have hiatuses because some lava flows were unsuitable for sampling (too fragile), whereas some segments of the coastal outcrops were inaccessible.

## COMPARISON OF THE OBTAINED PALEOMAGNETIC RESULTS WITH THE DATA ON THE OLD RED SANDSTONE (ORS) FORMATION OF BRITISH CALEDONIDES

As was noted, the difficulties (inconsistencies) in the Devonian paleomagnetic record are a global problem. The ORS formation of British Caledonides, which are largely represented by lava flows, is probably one of the paleomagnetically best studied objects that are close in age and in the formation aspects to the sections of the Minusa troughs. The paleomagnetic

**Table 3.** Paleomagnetic directions contained in volcanics and sedimentary layers of Tuba, Koksa, Bellyk, Novoselovo, Sisim, and Truba sections (site means)

| No. | Cross section | Cluster | Site | SLONG | SLAT | *n* | Dg | Ig | Ds | Is | *k* | a95 | *T*1 | *T*2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Tuba | NWSE | 028 | 92.821 | 53.831 | 10 | 315.7 | −2.9 | 317.0 | −2.2 | 419.3 | 2.4 | 400–590 | 685–700 |
| 2 | | NWSE | 029 | 92.824 | 53.830 | 10 | 314.8 | 7.8 | 310.3 | 10.5 | 80.3 | 5.4 | 450–640 | 685–700 |
| 3 | | NWSE | 030 | 92.797 | 53.853 | 10 | 318.1 | 27.6 | 310.7 | 10.5 | 203.7 | 3.4 | 400–620 | 695–700 |
| 4 | Koksa (south) | S | 004 | 91.411 | 54.183 | 2 + 1c | 147.2 | 48.4 | 148.0 | 32.4 | 194.0 | 10.4 | 440–665 | 590–710 |
| 5 | | type-I NW | 005 | 91.411 | 54.183 | 5 + 1c | 157.5 | −19.6 | 158.5 | −35.5 | 113.7 | 6.4 | 590–665 | 710 |
| 6 | | N | 006sed | 91.411 | 54.183 | 4 | 341.9 | −41.2 | 340.1 | −25.5 | 17.5 | 22.6 | 380–620 | 710 |
| 7 | | SW | 007 | 91.411 | 54.183 | 5 | 225.6 | −35.9 | 237.7 | −38.5 | 53.2 | 10.6 | 380–650 | 700–710 |
| 8 | | SW | 008 | 91.411 | 54.183 | 4 + 3c | 228.1 | −22.4 | 235.1 | −25.0 | 23.3 | 13.3 | 520–650 | 690–710 |
| 9 | | SW | 009 | 91.410 | 54.185 | 5 + 1c | 237.3 | −37.0 | 249.3 | −36.3 | 53.2 | 9.4 | 380–680 | 700–710 |
| 10 | | P | 010 | 91.409 | 54.188 | 4 | 339.3 | 59.5 | 347.6 | 75.2 | 40.8 | 14.6 | 380–680 | 620–710 |
| 11 | | P | 011 | 91.409 | 54.188 | 6 | 337.6 | 60.3 | 344.6 | 76.1 | 37.6 | 11.1 | 480–690 | 680–710 |
| 12 | | NW | 012sed | 91.409 | 54.188 | 6 | 314.1 | 38.0 | 307.0 | 49.2 | 48.0 | 9.8 | 480–590 | 710 |
| 13 | | S | 013 | 91.407 | 54.190 | 6 | 180.4 | 16.4 | 179.1 | 2.4 | 317.4 | 3.8 | 380–620 | 710 |
| 14 | Koksa (north) | N | 070–1 | 91.398 | 54.208 | 6 | 1.2 | −22.3 | 353.7 | −37.6 | 71.7 | 8.0 | 350–560 | 685–700 |
| 15 | | N | 070–2 | 91.398 | 54.208 | 6 | 9.7 | −21.2 | 3.7 | −38.4 | 476.3 | 3.1 | 400–490 | 685–695 |
| 16 | | N | 070–3 | 91.398 | 54.208 | 6 | 3.5 | −18.9 | 357.3 | −34.9 | 72.7 | 7.9 | 300–530 | 660–700 |
| 17 | | N | 070–4 | 91.398 | 54.208 | 5 | 9.1 | −12.8 | 5.1 | −30.0 | 118.2 | 7.1 | 490–560 | 685–695 |
| 18 | | N | 070–5 | 91.398 | 54.208 | 5 | 2.5 | −14.9 | 357.2 | −30.7 | 92.9 | 8.0 | 450–620 | 685–695 |
| 19 | | P | 070–6 | 91.398 | 54.208 | 6 | 226.1 | 63.9 | 249.1 | 83.1 | 298.2 | 3.9 | 530–640 | 685–695 |
| 20 | | S | 070–7 | 91.398 | 54.208 | 6 | 196.0 | 23.8 | 190.5 | 42.0 | 105.4 | 6.6 | 350–620 | 685–695 |
| 21 | Bellyk | N | 017 | 91.298 | 54.479 | 5 + 5c | 317.3 | −42.4 | 319.1 | −25.2 | 54.0 | 6.8 | 585–690 | 653–710 |
| 22 | | N | 018sed | 91.296 | 54.480 | 6 | 336.5 | −73.6 | 334.9 | −58.6 | 104.2 | 6.6 | 120–604 | 400–653 |
| 23 | | S | 019 | 91.296 | 54.481 | 6 | 176.5 | 43.8 | 172.4 | 29.8 | 83.8 | 7.4 | 400–513 | 585–653 |
| 24 | | S | 020 | 91.294 | 54.480 | 1 + 3c | 182.3 | 39.1 | 177.9 | 25.7 | 55.0 | 17.0 | 342 | 564 |
| 25 | | S | 021 | 91.294 | 54.480 | 6 | 163.3 | 45.2 | 161.4 | 30.4 | 49.3 | 9.6 | 513–585 | 653 |
| 26 | | S | 022 | 91.287 | 54.486 | 6 | 202.7 | 79.8 | 173.2 | 67.1 | 164.4 | 5.2 | 342–513 | 540–653 |
| 27 | | S | 023 | 91.286 | 54.486 | 5 | 154.2 | 39.9 | 154.0 | 24.9 | 58.6 | 10.1 | 400–540 | 585–653 |
| 28 | | NWSE | 024 | 91.223 | 54.508 | 5 | 316.1 | −12.0 | 316.5 | 2.4 | 33.6 | 13.4 | 400–513 | 540–638 |
| 29 | Novoselovo-1 | P | 001 | 91.006 | 54.942 | 6 | 150.4 | 72.9 | 97.0 | 81.8 | 65.5 | 8.3 | 250–470 | 650–700 |
| 30 | | NWSE | 002 | 91.017 | 54.950 | 5 | 306.9 | 40.7 | 318.4 | 8.2 | 129.3 | 6.8 | 400–670 | 700 |
| 31 | | N | 002sed | 91.017 | 54.950 | 5 + 5c | 332.2 | −1.9 | 325.2 | −39.9 | 28.2 | 9.5 | 540–638 | 622–680 |
| 32 | | N | 003 | 91.021 | 54.951 | 3 | 338.5 | −34.8 | 328.4 | −60.4 | 87.0 | 13.3 | 200–300 | 400–700 |
| 33 | | S | 004 | 91.024 | 54.953 | 6 | 197.0 | 17.2 | 209.9 | 49.7 | 106.6 | 6.5 | 470–600 | 650–700 |
| 34 | | N | 005 | 91.025 | 54.955 | 3 | 336.0 | 0.7 | 332.3 | −34.5 | 22.3 | 26.7 | 400–440 | 530–560 |
| 35 | | NW | 006 | 91.024 | 54.957 | 5 + 2c | 281.6 | 58.8 | 321.2 | 36.1 | 24.2 | 12.8 | 400–650 | 560–700 |
| 36 | Novoselovo-2 | N | 015–1 | 91.142 | 54.074 | 3 + 3c | 306.1 | −46.9 | 318.9 | −46.3 | 91.8 | 7.5 | 350–670 | 530–700 |
| 37 | | N | 015–2 | 91.142 | 54.074 | 6 | 306.3 | −45.6 | 318.5 | −45.1 | 313.0 | 3.8 | 250–300 | 530–700 |
| 38 | | E | 016 | 91.168 | 54.076 | 7 | 117.4 | −42.0 | 108.2 | −57.2 | 29.2 | 11.4 | 300–600 | 580–700 |
| 39 | | E | 017 | 91.163 | 54.074 | 6 | 98.0 | −29.9 | 89.1 | −41.7 | 28.6 | 12.7 | 350–630 | 670–685 |
| 40 | | E | 018 | 91.164 | 54.074 | 5 | 107.2 | −23.5 | 101.4 | −37.3 | 23.3 | 16.2 | 470 | 560–630 |
| 41 | | E | 019 | 91.166 | 54.075 | 5 | 119.0 | −18.5 | 115.8 | −34.2 | 39.5 | 12.3 | 470–530 | 580–700 |

**Table 3.** (Contd.)

| No. | Cross section | Cluster | Site | SLONG | SLAT | *n* | Dg | Ig | Ds | Is | *k* | a95 | *T*1 | *T*2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 42 | Sisim | type-I E | 039 | 91.811 | 55.036 | 4 | 275.5 | 38.5 | 293.0 | 28.1 | 35.2 | 15.7 | 540–638 | 667–692 |
| 43 | | type-I E | 040 | 91.817 | 55.037 | 3 + 1c | 264.9 | 25.3 | 277.1 | 21.3 | 133.2 | 8.5 | 445–638 | 653–680 |
| 44 | | type-I E | 041 | 91.817 | 55.039 | 6 | 244.3 | 27.3 | 260.6 | 32.6 | 277.0 | 4.0 | 342–604 | 653–692 |
| 45 | | P | 043 | 91.812 | 55.045 | 4 | 190.3 | 66.5 | 301.5 | 81.3 | 28.1 | 17.6 | 513–638 | 653–680 |
| 46 | | type-I | 044 | 91.813 | 55.048 | 5 | 87.6 | 0.0 | 87.1 | 2.1 | 64.1 | 9.6 | 513–604 | 638–692 |
| 47 | | N | 045 | 91.820 | 55.052 | 6 | 10.3 | −17.0 | 16.0 | −43.3 | 19.7 | 15.5 | 445 | 680–692 |
| 48 | | NW | 049 | 91.846 | 55.068 | 2 + 4c | 303.9 | 66.8 | 319.7 | 41.9 | 103.7 | 7.4 | 400–482 | 585 |
| 49 | | NW | 050 | 91.844 | 55.068 | 6 | 283.9 | 67.7 | 310.4 | 45.9 | 262.8 | 4.1 | 260–482 | 513–585 |
| 50 | | S | 052 | 91.844 | 55.071 | 6 | 143.2 | 42.9 | 140.0 | 61.8 | 61.5 | 8.6 | 400–513 | 585–692 |
| 51 | | type-I | 053 | 91.844 | 55.072 | 6 | 30.6 | 7.4 | 29.8 | −1.8 | 286.7 | 4.0 | 342–513 | 564–653 |
| 52 | | N | 054 | 91.837 | 55.077 | 1 + 5c | 359.1 | −33.3 | 8.7 | −49.0 | 33.4 | 13.9 | 564 | 692 |
| 53 | | N | 057 | 91.839 | 55.076 | 5 + 1c | 317.6 | −47.8 | 309.8 | −66.1 | 83.9 | 7.5 | 260–482 | 513–653 |
| 54 | Truba | S | 030 | 92.354 | 55.483 | 5 | 153.0 | 51.1 | 167.5 | 39.8 | 71.6 | 9.1 | 400–622 | 638–653 |
| 55 | | S | 031 | 92.365 | 55.487 | 6 | 170.7 | 46.3 | 179.6 | 31.8 | 90.5 | 7.1 | 445 | 653 |
| 56 | | S | 032 | 92.370 | 55.491 | 6 | 162.6 | 43.9 | 172.4 | 31.0 | 84.4 | 7.3 | 445–564 | 653 |
| 57 | | S | 033 | 92.380 | 55.493 | 6 | 185.3 | 40.4 | 190.0 | 24.0 | 73.7 | 7.9 | 445–513 | 653 |
| 58 | | S | 033–2 | 92.380 | 55.493 | 6 | 186.4 | 48.3 | 192.3 | 31.6 | 121.1 | 6.1 | 445–564 | 653 |
| 59 | | S | 033–3 | 92.380 | 55.493 | 6 | 172.0 | 38.4 | 178.6 | 23.9 | 252.5 | 4.2 | 400 | 653 |
| 60 | | S | 033–4 | 92.380 | 55.493 | 5 | 174.3 | 45.0 | 182.1 | 29.9 | 129.9 | 6.7 | 400–445 | 585–653 |
| 61 | | S | 034 | 92.382 | 55.494 | 6 | 175.0 | 48.3 | 183.5 | 33.1 | 419.2 | 3.3 | 400–445 | 653 |
| 62 | | NWSE | 035 | 92.413 | 55.519 | 3 + 3c | 128.4 | 8.3 | 130.7 | 5.9 | 39.4 | 11.6 | 445–513 | 564–653 |
| 63 | | S | 036 | 92.414 | 55.520 | 3 + 3c | 166.5 | 55.9 | 179.6 | 41.8 | 17.0 | 17.8 | 445–540 | 638 |
| 64 | | S | 039 | 92.416 | 55.522 | 6 | 128.4 | 51.6 | 148.5 | 46.4 | 17.7 | 16.4 | 120–482 | 513–540 |
| 65 | | S | 040 | 92.418 | 55.524 | 5 | 141.7 | 46.7 | 156.5 | 38.3 | 12.8 | 22.2 | 400 | 540–585 |
| 66 | | S | 041 | 92.418 | 55.524 | 4 | 132.7 | 66.6 | 164.2 | 58.2 | 16.1 | 23.6 | 120–445 | 540 |
| 67 | | S | 073 | 92.360 | 55.484 | 6 | 164.3 | 40.7 | 172.8 | 27.5 | 140.6 | 5.7 | 241–482 | 482–680 |
| 68 | | S | 074 | 92.418 | 55.524 | 5 | 194.0 | 80.3 | 190.1 | 38.4 | 31.0 | 14.0 | 400–513 | 540–680 |

Cross-section is section name; cluster is cluster to which direction from respective site pertains. Index "sed" in column "site" indicates data from sedimentary rocks. For type I directions (accidental directions), in column "cluster," belonging of direction to respective cluster was determined by repeated interpretation (see Section *Data Synthesis*). Other explanations are in Tables 1 and 2 and in text.

anomalies similar to the Minusa ones were also recorded in these objects. In particular, the analysis of the paleomagnetic data from the ORS formation of the British and Scandinavian Caledonides which include, in addition to the determinations from the stratified complexes, also the determinations from intrusions and subvolcanic formations, allowed Piper (2007) to conclude on the existence of the Devonian episode of the true polar wander (TPW). In fact, it was an attempt to explain the significant discrepancies between the Devonian determinations by the phenomena of global tectonics. It should be noted here that the determinations forming the Devonian loop on the APWP curve which, according to Piper, correspond to the TPW event (Piper, 2007), were obtained from the intrusive complexes, i.e., from the objects for which it is difficult and often impossible to reconstruct their primary bedding. Naturally, this makes Piper's hypothesis quite vulnerable.

We made an attempt to more strictly analyze the paleomagnetic data for the ORS formation in order to compare them to the Minusa results. Just as in the case with the Minusa objects, we carried out the cluster analysis of the data for individual sites aimed at revealing the predominant directions. The analysis was conducted exclusively for the determinations obtained from the stratified volcanic rocks because, on one hand, these objects allow clear and reliable identification of the primary bedding and, on the other hand, the volcanics are free of the effect of inclination shallowing, which is typical of sedimentary rocks. We analyzed the data presented in (Sallomy and Piper, 1973; Latham and Briden, 1975; Kono, 1979; Torsvik, 1985) for the Midland Valley and Lorne Plateau lavas, Scotland. Other publications on the subject only contain averaged determinations. The primary selection criterion of the data was based on the statistical parameters: the final dataset was compiled of the directions obtained from at least two samples and the confidence

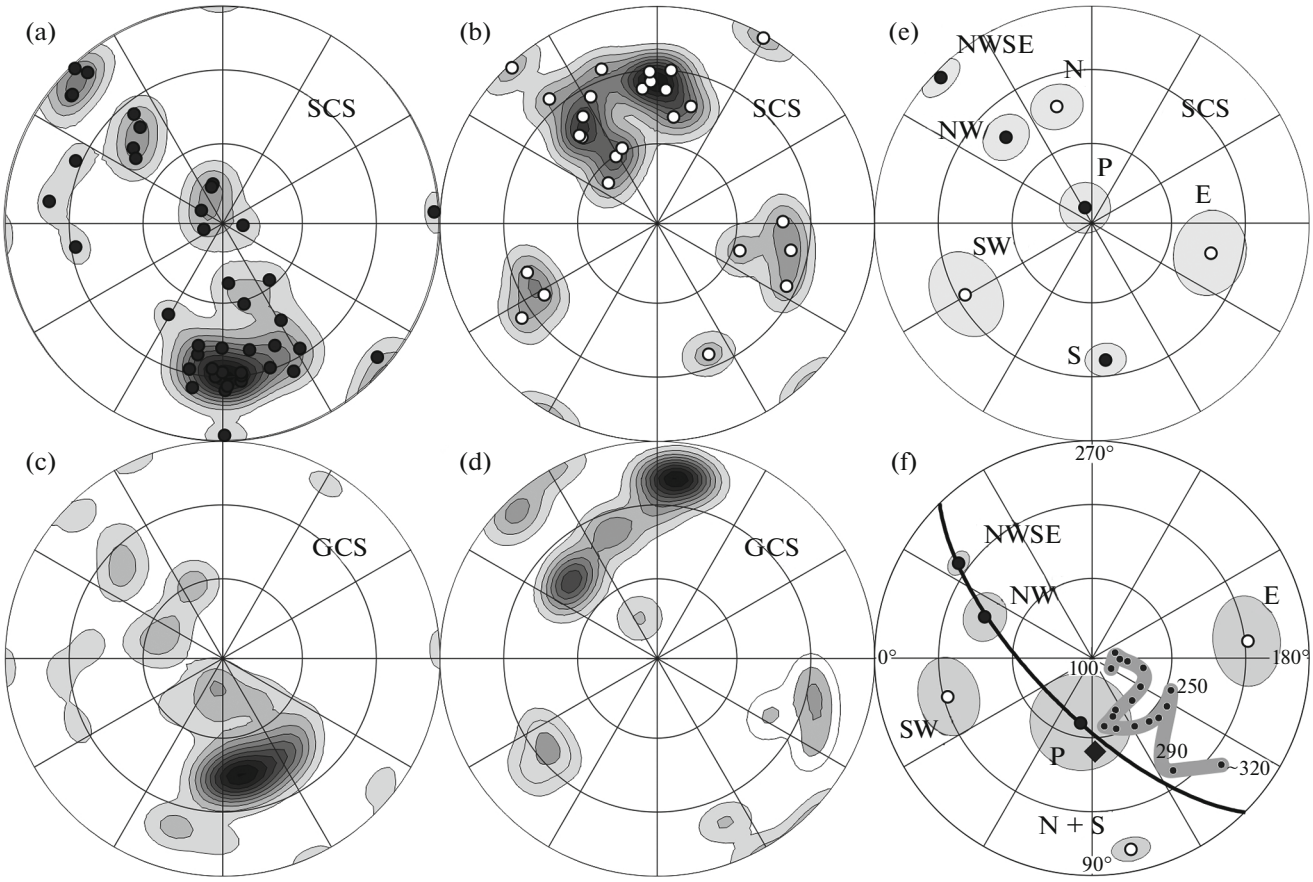



**Fig. 5.** (a)–(d) Cluster analysis of site directions, (e) cluster-averaged paleomagnetic directions, and (f) paleomagnetic poles for clusters from Devonian rocks of Tuba, Koksa, Bellyk, Novoselovo, Sisim and Truba zections. (a), (c), distribution density of site directions with positive inclinations (lower hemisphere); (b), (d) distribution density of site directions with negative inclinations (upper hemisphere). Diagram (f): filled/open markers are northern/southern hemisphere; black line is great circle drawn through poles for clusters P, NW, NWSE; diamond is position of study area; heavy gray line and respective numbers are APWP curve for stable Europe for interval of 250–100 Ma (Torsvik and Cocks, 2005) combined with known Late Paleozoic poles of Siberia (Shatsillo et al., 2012; 2014). Other explanations are in Fig. 2 and in text.

radius was at most 20°. These criteria were met by the data from 58 sites (lava flows). The age of the Midland Valley and Lorne Plateau lavas covers the interval from 411 to 394 Ma (Browne et al., 2002); i.e., it is almost identical to the age of the Minusa volcanics.

The results of the cluster analysis for the directions recorded in the ORS lavas of Scotland are illustrated in Fig. 7 and presented in Table 6. As can be seen in Figs. 7a–7c, against the general quasi-chaotic distribution of the directions, statistically more or less substantiated clusters are present in the densitograms. An overwhelming number of the directions form cluster NE and the diffuse SW cluster which are close to antipodal. The comparison of the mean directions over these clusters reduced to the same polarity indicates the absence of statistically significant distinctions between them ($\gamma/\gamma_c$ = 10.3/12.3). The NE and SW directions are the expected ones for the Devonian of Scotland; in particular, the paleomagnetic pole calculated from these directions (Table 6) is statistically indistinguishable from the Early Devonian pole from Podolia (PLong = 145.5; Plat = –3.7; A95 = 6.7) (Smethurst and Khramov, 1992), which is considered as the reference for the East European platform. Furthermore, also the NEE cluster and two statistically poorly substantiated NW and E clusters are quite clearly identified. Just as the Minusa objects, the OSR lavas, in addition to the predominant groups, also contain separate accidental directions which do not fit the general distribution.

The paleomagnetic poles of ORS lavas calculated from all the clusters significantly differ from the known post-Devonian European poles; i.e., it is problematic to interpret the respective directions in the context of the hypothesis of post-Devonian remagnetization.

The distribution of the ORS poles by clusters is confidently approximated by the great-circle arc (Fig. 7d); moreover, the virtual geomagnetic poles (VGPs) (from the sites) do not constitute a directed trend in accor-

**Table 4.** Cluster-mean paleomagnetic directions and poles recorded in Devonian rocks of Tuba, Koksa, Bellyk, Novoselovo, Sisim, and Truba sections

| Type | Cluster | *n* | Dg | Ig | Ds | Is | *k* | a95 | FT(DC) | PLong | PLat | *k* | A95 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| II | SW | 3 | 230.2 | –31.9 | 240.5 | –33.4 | 71.3 | 14.7 | x | 15.0 | –31.8 | 90.5 | 13.0 |
| | E | 4 | 110.8 | –28.5 | 104.2 | –42.9 | 40.0 | 14.7 | x | 186.3 | –28.7 | 40.5 | 14.6 |
| III | NWSE | 6 | 313.5 | 8.6 | | | 14.4 | 18.3 | + | | | | |
| | | | | | 313.9 | 3.9 | 105.9 | 6.5 | | 324.6 | 25.5 | 250.4 | 4.2 |
| | P | 5 | 268.1 | 82.3 | | | 8.7 | 27.4 | + | | | | |
| | | | | | 336.4 | 83.7 | 65.3 | 9.5 | | 79.8 | 65.3 | 17.9 | 18.6 |
| | NW | 4 | 298.6 | 58.7 | | | 24.8 | 18.8 | + | | | | |
| | | | | | 315.0 | 43.4 | 115.3 | 8.6 | | 339.0 | 46.0 | 113.9 | 8.6 |
| | S | 24 | 168.9 | 47.3 | | | 16.6 | 7.5 | ? | | | | |
| | | | | | 174.2 | 36.8 | 19.3 | 6.9 | | | | | |
| | N | 16 | 345.8 | –31.3 | | | 9.1 | 13.0 | + | | | | |
| | | | | | 343.3 | –43.3 | 17.5 | 9.1 | | | | | |
| MEAN N + S | | 40 | 167.5 | 41.1 | | | 11.5 | 7.0 | ? | | | | |
| | | | | | 170.1 | 39.5 | 17.7 | 5.5 | | 101.5 | –11.1 | 16.5 | 5.7 |

Type is type of paleomagnetic direction and its occurrence in sections: II, directions that occur in only one section; III, directions that occur in remote sections; *n* is number of sites; FT(DC) is result of Direction-Correction fold test (Enkin, 2003) (x indicates that test is inapplicable because of absence of variations in attitude of bedding; ? indicates that result is uncertain; + indicates that magnetization is pre-folding); PLong/Plat are longitude/latitude of paleomagnetic pole; A95 is radius of confidence circle corresponding to probability of 95%. Poles are calculated for coordinates 54.6° N, 91.7° E. Other explanations are in Table 1 and in text.

**Table 5.** Ratio of paleomagnetic directions revealed in Devonian rocks of Tuba, Koksa, Bellyk, Novoselovo, Sisim, and Truba sections in accordance with stratigraphic sequence (cluster/site number)

| Tuba | Koksa (south) | Koksa (north) | Bellyk | Novoselovo-1 | Novoselovo-2 | Sisim | Truba |
|---|---|---|---|---|---|---|---|
| NWSE/030 | S/013 | S/070-7 | N/017 | NW/006 | E/016 | N/054 | S/030 |
| NWSE/029 | NW/012 | P/070-6 | N/018 | N/005 | E/019 | N/057 | S/073 |
| NWSE/028 | P/011 | N/070-5 | S/019 | S/004 | E/018 | type-I/053 | S/031 |
| | P/010 | N/070-4 | S/020 | N/003 | E/017 | S/052 | S/032 |
| | SW/009 | N/070-3 | S/021 | N/002sed | N/015-1 | NW/050 | S/033 |
| | SW/008 | N/070-2 | S/022 | NWSE/002 | N/015-2 | NW/049 | S/033-2 |
| | SW/007 | N/070-1 | S/023 | P/001 | | N/045 | S/033-3 |
| | N/006 | | NWSE/024 | | | type-I/044 | S/033-4 |
| | type-I NW/005 | | | | | P/043 | S/034 |
| | S/004 | | | | | type-I E/041 | NWSE/035 |
| | | | | | | type-I E/040 | S/036 |
| | | | | | | type-I E/039 | S/039 |
| | | | | | | | S/040 |
| | | | | | | | S/074 |
| | | | | | | | S/041 |

Bottom of columns corresponds to base of studied fragment of section. Other explanations are in Table 3.

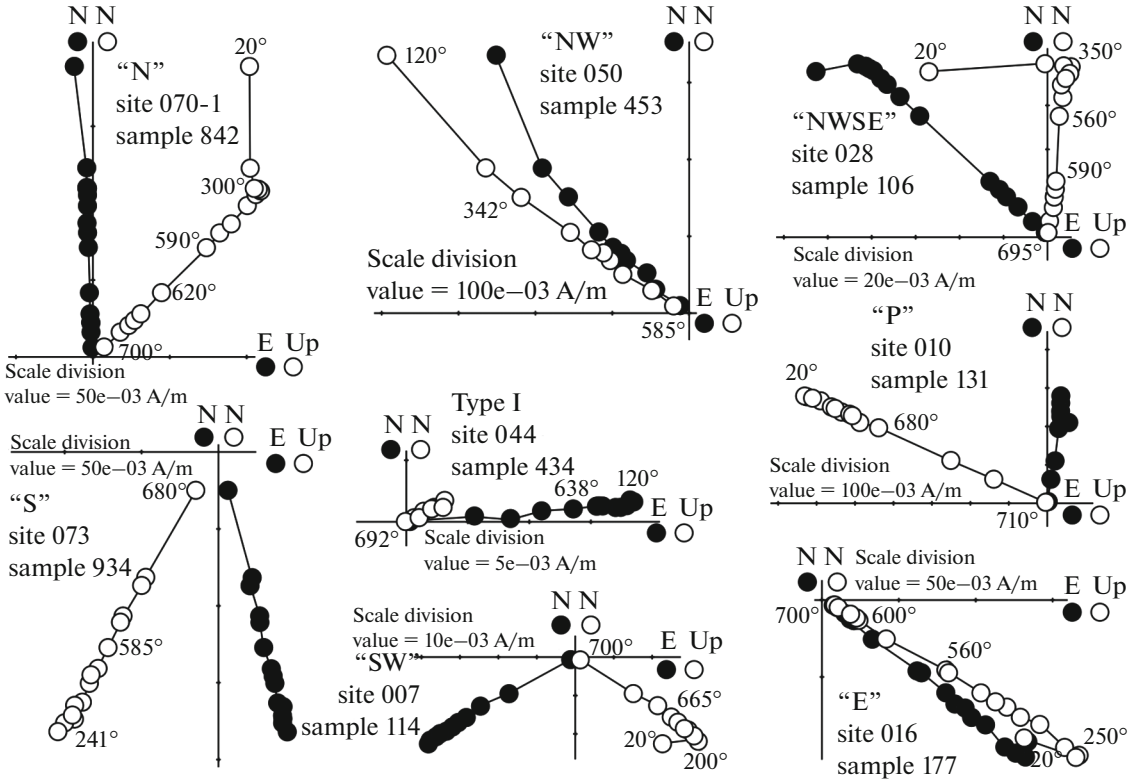



**Fig. 6.** Examples of thermal demagnetization of Devonian volcanics characterizing clusters revealed from Tuba, Koksa, Bellyk, Novoselovo, Sisim, and Truba sections. Zijderveld diagrams are presented in stratigraphic coordinate system.

dance with the stratigraphic sequence in the sections. The complicated reciprocating movements of the VGPs are poorly consistent with the true polar wander hypothesis which is used in (Piper, 2007) for explaining the distribution of the Devonian paleomagnetic directions for Europe.

The results of our analysis also indicate that the paleomagnetic determinations for the Devonian obtained from the large intrusive bodies ((Piper, 2007) and references there) whose cooling (in contrast to the lava flows) took a fairly long time are barely suitable for solving the geophysical problems. This is due to the fact that the paleomagnetic directions recorded in these intrusions *can* be the superposition of several differently directed magnetization components that are close to each other in age.

### *Data Synthesis: Chaos or Regularity?*

It was previously shown (Shatsillo, 2015a) that the Early Permian paleomagnetic poles for Siberia and the East European platform (Baltica) can be brought in coincidence by the rotation around the Euler pole

**Table 6.** Cluster-mean paleomagnetic directions and poles for Devonian volcanics of ORS formation, Scotland

| Cluster | *n* | Ds | Is | *k* | a95 | PLong | PLat | *k* | A95 | PLong(R) | PLat(R) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| E | 3 | 98.4 | 46.1 | 64.1 | 15.5 | 64.3 | 18.2 | 48.1 | 18.0 | 21.1 | 25.0 |
| NE | 26 | 46.6 | –45.7 | 25.3 | 5.7 | | | | | | |
| NEE | 8 | 67.9 | 8.3 | 30.2 | 10.2 | 103.5 | 15.6 | 36.2 | 9.3 | 60.7 | 17.5 |
| NW | 3 | 328.3 | –49.6 | 35.4 | 21.0 | 203.7 | –1.7 | 30.0 | 22.9 | 158.4 | –10.2 |
| SW | 16 | 218.6 | 37.3 | 8.9 | 13.1 | | | | | | |
| SW* | 9 | 216.0 | 42.9 | 21.8 | 11.3 | | | | | | |
| NE + SW* | 35 | 43.8 | –45.1 | 24.0 | 5.1 | 139.3 | –2.3 | 21.8 | 5.3 | 93.5 | –5.4 |

* Mean direction for cluster SW was calculated based on site directions falling in center of distribution in densitogram (Fig. 7a). Poles are calculated for coordinates 56.5° N, 357.0° E; PLong(R)/PLat(R) are longitude and latitude of paleomagnetic pole reduced to Siberian coordinate system (explanations are in Fig. 8); other explanations are in Tables 1, 4 and in text.

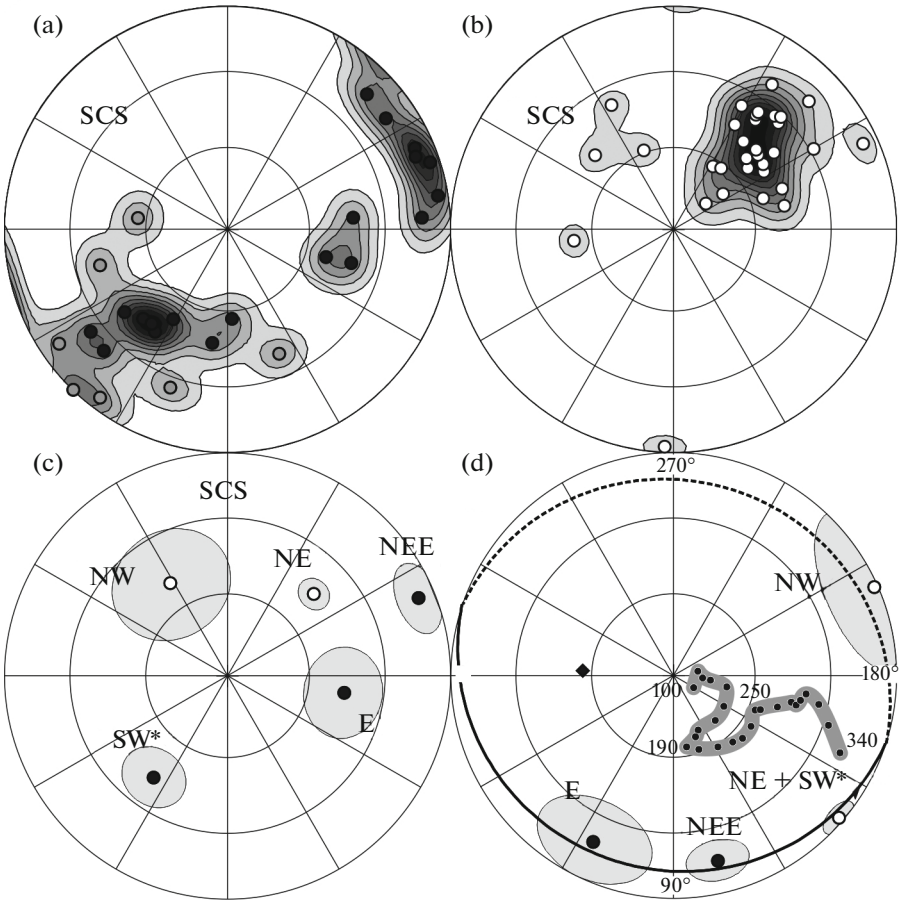



**Fig. 7.** (a), (b) Cluster analysis of site-mean paleomagnetic directions; (c) cluster-averaged paleomagnetic directions, and (d) paleomagnetic poles for clusters for Devonian ORS volcanics, Scotland. (a) Distribution density of site-mean directions with positive inclinations (lower hemisphere); (b) distribution density of site-mean directions with negative inclinations (upper hemisphere); in diagram (d), filled/open markers are northern/southern hemisphere; black line is great circle drawn through poles for clusters; diamond is geographic position of ORS formation; gray line in bold and respective numbers are APWP curve for stable Europe for interval of 340–100 Ma (Torsvik and Cocks, 2005). Other explanations are in Fig. 2 and in text. * Cluster-mean direction was calculated based on site-mean directions falling in center of distribution in densitogram (a).

located within the Kara Sea, which retrospectively describes the character of the interaction between the cratons during their Permian consolidation within the structure of Eurasia. This pattern of the interaction between Siberia and Baltica is consistent with a number of the specific Late Paleozoic–Early Mesozoic structures in the folded regions surrounding them and does not conflict with the kinematic constraints of the recent plate tectonics.

The analysis of the most recent paleomagnetic data on the Paleozoic of Siberia and Baltica (Shatsillo, 2015b; Shatsillo et al., 2017) has shown that these continental blocks could have been in the Early Permian configuration since the Silurian, i.e., from the Silurian to the beginning of the Permian, they were a single lithospheric plate that underwent reorganization during the Permian. Thus, using the Euler pole suggested in (Shatsillo, 2015a), we can bring the paleomagnetic poles for the Devonian of Caledonian Siberia and European Caledonides to their initial mutual position within a single plate. The ORS poles reduced in this way to the Siberian coordinates are presented in Table 6 and shown in Fig. 8. As can be seen in Fig. 8a, as a result of this reconstruction, the poles for the expected Devonian directions (N + S and NE + SW) became coincident, in full correspondence with what should follow from the model (Shatsillo, 2015b).

At the same time, the obvious convergence of the poles corresponding to the anomalous directions is absent. The coincidence can be achieved for pairs of poles (E and E) and (NWSE and NW) but only if the polarity of one of the directions is changed. This variant of the interpretation needs some substantiation because the absolute predominance of monopolar directions in each of the studied regions which is revealed by the cluster analysis does not give grounds for these assumptions: it is required that at least single antipodal vectors exist for each cluster in each region. For checking the existence of these antipodal vectors, let us again return to the analysis of the distribution of the

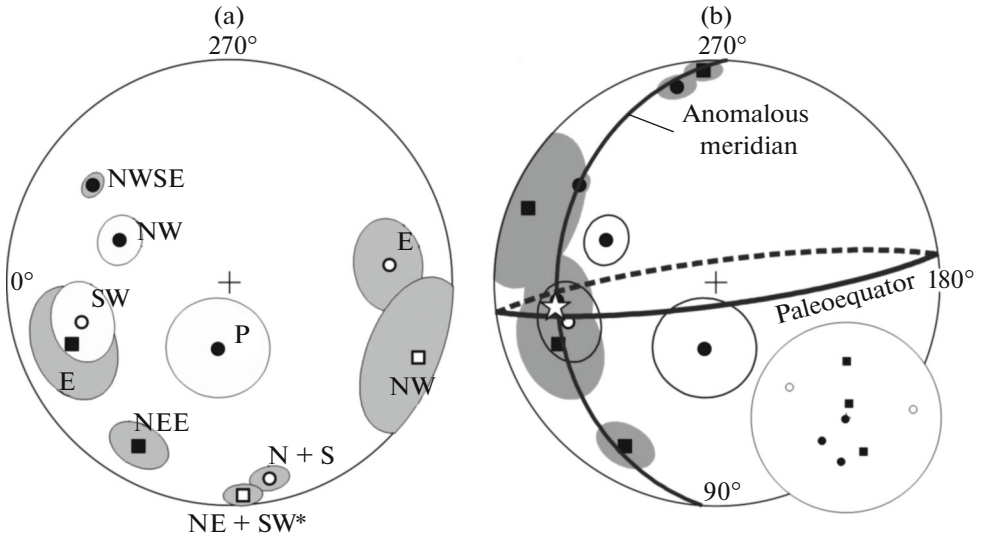



**Fig. 8.** Comparison of calculated cluster paleomagnetic (?) poles for Devonian rocks of Scotland and Minusa trough reduced to Siberian coordinate system. Squares are poles for objects of Scotland; circles are poles for Minusa objects; filled and open markers correspond to northern and southern hemisphere, respectively; asterisk is mean pole over all anomalous clusters. Confidence circles filled with gray are for poles based on which approximating circle (paleomeridian) is built. Inset shows distribution of anomalous poles reduced to same polarity and to center of the sphere (open and filled markers for monopolar and bipolar clusters, respectively). Scotland poles are reduced to Siberian coordinate system by clockwise rotation by 46° around pole with coordinates 79° N, 95° E (according to (Shatsillo, 2015a)). Other explanations are in text.

single directions that were classified as *accidental* during the identification of the general clusters (Figs. 5, 7).

*The Minusa objects*. (1) In the NWSE cluster, we initially included a single direction with an SE declination (site 035, the Truba section). This was done based on the presence of high-temperature final and medium-high-temperature intermediate magnetization components, antipodal to each other, recorded in the samples of this site; a similar paleomagnetic record was also observed in the samples from site 029 of the Tuba section. (2) As follows from Figs. 5a, 5b, and 5e for cluster E, we see a close-to-antipodal "diffuse" cluster (and therefore not considered in the statistics) with WNW declinations, which includes the directions from three sites of the Sisim section (039, 040, and 041). The expanded (due to the directions from the Sisim section) cluster E is determined by the fold tests as prefolding. (3) For cluster NW, also, a close-to-antipodal single direction with an SE declination is observed.

For all the anomalous clusters identified in the *objects from Scotland* (NEE, NW, and E), the stereograms contain antipodal directions (Figs. 7a–7c). Three directions that are close-to-antipodal to the NEE cluster fall in the extreme SW part of the distribution for the expected Devonian SW direction (these directions were not considered in the calculation of the common mean for the SW and NE clusters). Two single directions with the SE and W declinations are antipodal to clusters NW and E, respectively.

Thus, the vast majority of the directions identified by cluster analysis in the objects from Minusa and Scotland can be considered bipolar. The Minusa clusters P and SW are the only exception. Now, for analyzing the entire set of the anomalous poles let us reduce them to one polarity. The character of the resulting distribution is shown in Fig. 8b. It can be seen that all the bipolar anomalous poles are fairly accurately approximated by the arc of the great circle passing through the expected Devonian poles; i.e., they lie along the paleomeridian, which we refer to as the anomalous meridian. This trend is a determinant in the geometry of the entire range of directions. The most soundly statistically substantiated anomalous poles (NEE for Scotland and NWSE for Minusa) are located at a distance of 45° from the intersection of the anomalous meridian and the paleoequator. The other bipolar anomalous poles are located between the NEE and NWSE poles and are closer to the paleoequator. The positions of the monopolar anomalous poles (the P and SW Minusa poles) also obey a certain regularity: with the confidence interval taken into account, these poles are close to the paleoequator and located at a distance of 50° to 60° from the anomalous meridian. The pole averaged over all the anomalous directions falls at the intersection of the paleoequator and the anomalous meridian. Thus, the distribution of the anomalous poles for Scotland and Minusa has a clear symmetry.

## CONCLUSIONS

The paleomagnetic data obtained from the Early and Middle Devonian objects of the Minusa troughs show that in the Devonian, at least during 10–20 Ma, the directions of the geomagnetic field were extremely variable (hyperactive) and were substantially different

in this respect from the geomagnetic field of most of the Phanerozoic. Here, against rapid variations of the directions, there are clusters of the paleomagnetic directions (poles) corresponding to the certain quasi-stable states of the field. A comparison of the obtained results with the previously published data for the Devonian of Scotland show that the observed features of the paleomagnetic record are not local, peculiar to a specific region but, rather, are likely to occur globally.

Perhaps, some analogies for this behavior are met by us in the paleomagnetic record of the Vendian–Cambrian boundary interval ((Shatsillo et al., 2005) and references there).

The closeness of a number of clusters of the paleomagnetic poles calculated for the south of Siberia and for Scotland (with tectonic corrections according to (Shatsillo, 2015b)) suggests that the geomagnetic field during the considered time interval preserved, at least partially, its predominant dipole geometry with an axis that could have significantly deviated from the axis of the Earth's rotation.

The position of most of the identified clusters along the great circle passing through the region of the location of the expected Devonian paleomagnetic pole (i.e., the pole that agrees with the general trend of the Siberian APWP curve) indicates that the clusters corresponding to the quasi-stable directions of the field were located along the paleomeridian. This can be interpreted as evidence of the significant contribution of the equatorial dipole to the geomagnetic field in the considered interval of the Devonian.

According to the results of (Shcherbakova et al., 2017), this time interval was also characterized by the extremely low intensity of the geomagnetic field, which is a direct indication of the weakening of the axial dipole compared to the field of the "ordinary" epochs. This weakening of the axial dipole, along with the existence of a relatively strong equatorial dipole, provided that both dipoles make a variable contribution to the total geomagnetic field, could explain the anomalous Devonian directions forming the cloud of poles which is elongated along the paleomeridian.

The contribution of the equatorial dipole could increase as the contribution of the axial dipole decreased (for example, during its reversals), which caused the approach of anomalous poles to the paleoequator. The possibility of the existence of a relatively strong and stable equatorial dipole whose contribution in the main magnetic field in the Paleozoic could be from 5 to 20% was previously noted in (Khramov, 2007; Khramov and Iosifidi, 2012).

The tectonic interpretation of the obtained data falls beyond the scope of this paper; however, we note two points.

(1) In the light of the obtained data and the analysis of the published results, the attempts to explain the significant discrepancy of the Devonian poles for a single block by the global tectonic phenomena (the TPW hypothesis) (Piper, 2007) seem invalid;

(2) The data for the "expected" Devonian directions for the European and Asian parts of Eurasia support the hypothesis (Shatsillo, 2015b; Shatsillo et al., 2017) of the stable mutual position of Siberia and Baltica during the Middle and Late Paleozoic.

The unusual character (relative to most of the Phanerozoic) of the Devonian geomagnetic field can indicate the existence of some peculiarities (inhomogeneities) at the core/mantle boundary which caused the anomalous behavior of the geodynamo in that epoch. The Devonian geomagnetic phenomenon needs further research and comprehension.

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